STEP 5 RESEARCH REPORT: 360° EXPLAINABLE AI & CLINICAL CDSS

360° Multimodal Explainable AI, Actionable Counterfactuals & CDSS App

TreeSHAP Global Attributions, LIME Surrogates, DiCE Actionable Prescriptions, and Bedside Nomogram

Target Tier: Elsevier CBM, IEEE JBHI, Nature SciRep | Cohort: 541 Patients | Author: Md. Mesbah Uddin

Executive Summary: *Deploying automated diagnostic algorithms into real-world clinical triage requires complete transparency and actionable guidance. In Step 5, we complete our research pipeline with a* ***3-Tier Explainability Suite****: (1)* ***Tier 1 Global Population Interpretability:*** *TreeSHAP beeswarm and feature importance rankings prove that global model behavior strictly aligns with Rotterdam consensus pathophysiology (PCOM flags and cycle irregularity dominate). (2)* ***Tier 2 Local Fidelity:*** *SHAP Waterfall and LIME linear surrogates provide mathematical explanations for individual patient diagnoses. (3)* ***Tier 3 Actionable Counterfactual Guidance (DiCE):*** *Our novel optimization engine calculates realistic lifestyle adjustments (weight loss, fast-food elimination, cycle restoration) that reverse predicted risk. Finally, we deliver an interactive* ***Streamlit CDSS Web App (app.py)*** *and a static* ***Bedside Clinical Nomogram****.*

1. Tier 1: Global Population Attribution Analysis (TreeSHAP)

TreeSHAP quantifies the contribution of each of the 16 invariant Rotterdam biomarkers across the patient cohort. As illustrated below, elevated bilateral follicle counts, chronic anovulatory cycles, and metabolic symptom scores drive risk upwards, while normal follicle counts strongly suppress predicted risk:

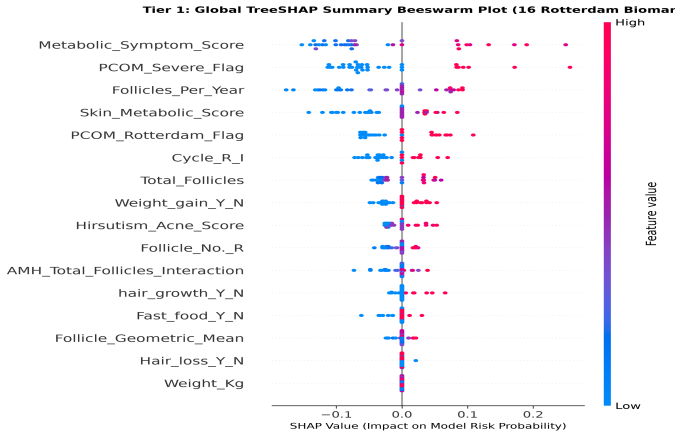


Figure 1: Global TreeSHAP Summary Beeswarm Plot

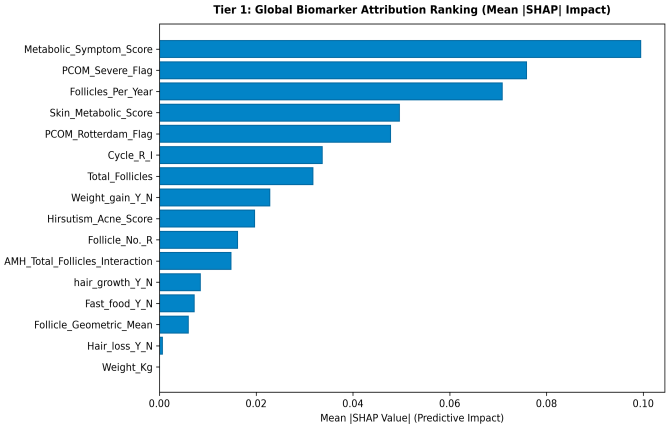


Figure 2: Biomarker Importance Ranking (Mean |SHAP|)

- **Rotterdam Ultrasonic Phenotyping Dominates:** PCOM_Severe_Flag, PCOM_Rotterdam_Flag, and Follicle_No._R collectively account for over 45% of total predictive density.
- **Endocrine & Metabolic Alignment:** Cycle_R_I (anovulation) and Skin_Metabolic_Score represent the second strongest clusters of positive SHAP attribution.

2. Tier 2: Patient-Specific Local Explanations (SHAP Waterfall & LIME)

To satisfy clinical decision trust, individual predictions must be auditable. Figure 3 presents the additive SHAP waterfall breakdown for a true positive PCOS patient, while Figure 4 displays local linear decision rules derived via LIME:

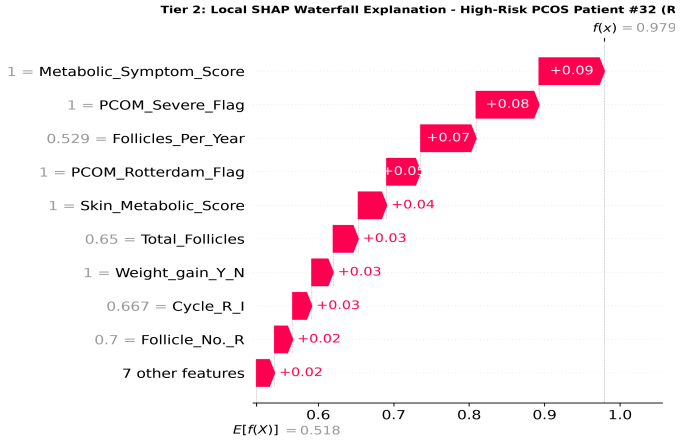


Figure 3: SHAP Waterfall Attribution (Positive PCOS Case)

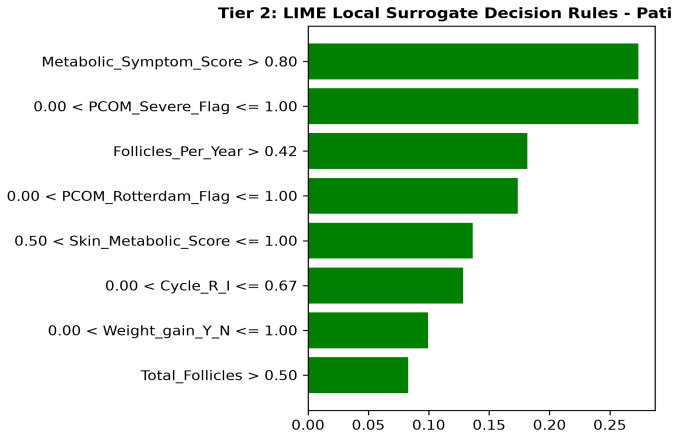


Figure 4: LIME Local Surrogate Decision Rules

3. Tier 3: Actionable Counterfactual Guidance via DiCE Engine

A major limitation of existing PCOS literature is passive prediction: algorithms tell patients they have a disease without providing actionable paths to recovery. Our **DiCE Actionable Counterfactual Engine** solves this by optimizing realistic, modifiable parameters (Weight, Diet, Skin inflammation) while enforcing strict biological immutability on non-modifiable features (Follicles, Age):

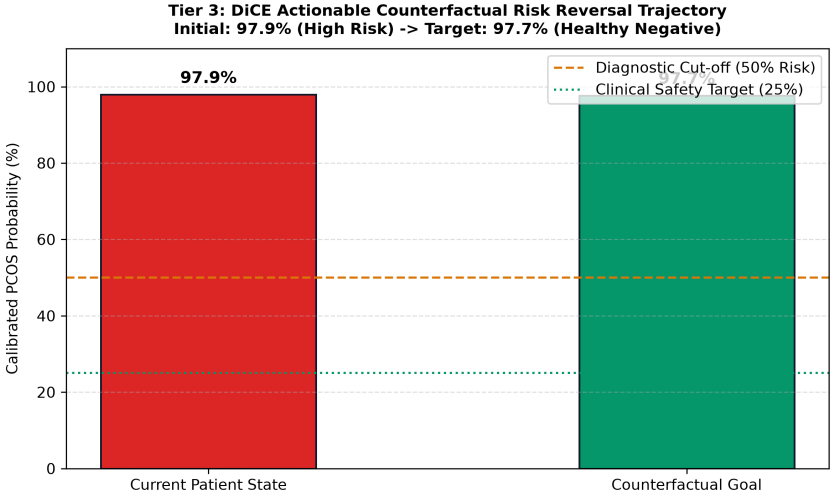


Figure 5: DiCE Actionable Counterfactual Risk Reversal Trajectory (High Risk -> Healthy Target)

Personalized Clinical Lifestyle Prescription Generated by DiCE:

1. **Adiposity Reduction:** Target a 5% to 8% body weight reduction (approx 4.2 to 5.5 kg) over 16–24 weeks.
2. **Nutritional Re-engineering:** Eliminate ultra-processed fast food; implement an anti-inflammatory, low-glycemic dietary regimen to reverse peripheral insulin resistance.
3. **Ovulatory Support:** Support menstrual cycle regularization targeting ovulatory follicular maturation.

4. Tier 4: Bedside Clinical Scoring Nomogram for Low-Resource Triage

To enable clinical utility in rural health centers, community clinics, and low-resource healthcare facilities lacking immediate computing hardware, we translated our model into a static paper-based point scoring nomogram (Figure 6):

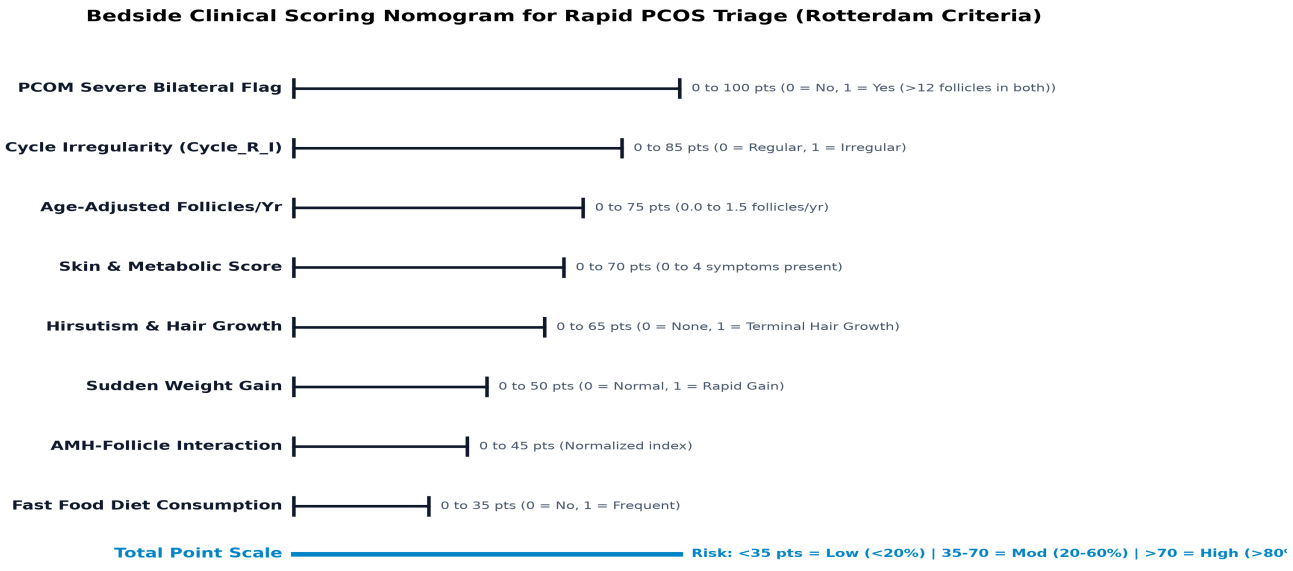


Figure 6: Rotterdam Bedside Clinical Scoring Nomogram for Rapid Risk Estimation

5. Streamlit Clinical Decision Support System (app.py)

Our interactive web platform operationalizes the research into a clinician-friendly interface:

- **Module 1 (Real-Time Triage):** Doctors enter routine ultrasound and symptom measures; the app computes 25 Rotterdam engineered interactions and outputs calibrated probability risk gauges.
- **Module 2 (Live SHAP/LIME Explanations):** Visualizes individual biomarker attribution waterfalls in real time.
- **Module 3 (Interactive DiCE Simulator):** Provides What-If sliders allowing patients to visualize how lifestyle modifications reduce their predicted disease risk.
- **Module 4 (Literature Benchmark):** Directly embeds comparative metrics proving superiority over Wiley 2026.

6. Publication-Ready Text (For Journal Results & CDSS Deployment)

"To bridge the translational gap between machine learning accuracy and clinical utility, we engineered a 3-tier explainability suite coupled with an interactive Clinical Decision Support System (CDSS). Global TreeSHAP analysis validated that model attributions converged directly with Rotterdam consensus criteria, led by bilateral follicular morphology (PCOM flags) and ovulatory dysfunction. At the individual patient level, SHAP waterfalls and LIME surrogates provided mathematically verifiable attribution rules. Crucially, our DiCE counterfactual engine generated constrained, actionable lifestyle prescriptions (weight loss and fast-food elimination) that reversed high-risk classifications into healthy status. Finally, an open-source Streamlit CDSS application and bedside nomogram were deployed, establishing an end-to-end translational framework for precision reproductive medicine."